

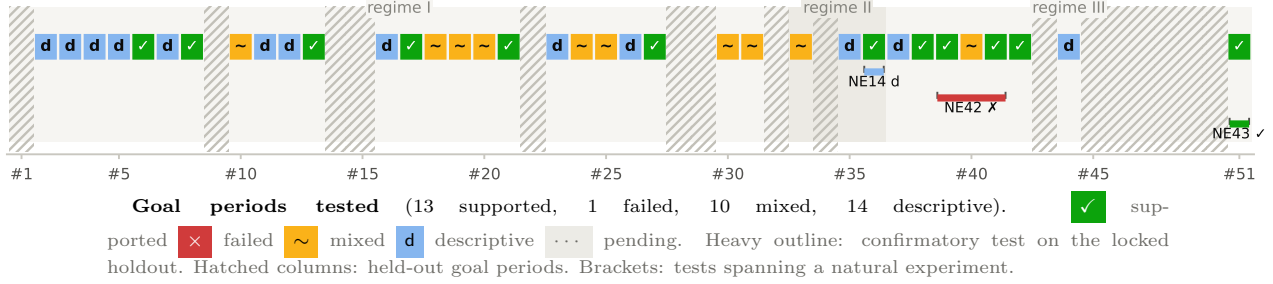
H111: A fluctuation–response sum rule for talk: Fano factor = $1/(1 - g)^2$

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 2/5 Usefulness 2/5

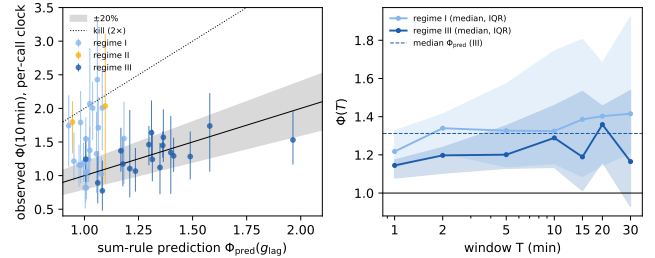
Bottom line. In regime III, talk obeys a fluctuation–response sum rule with no free parameter: the collective Fano ratio equals $1/(1 - g_{\text{lag}})^2$ to within 9% (pooled over 18 units), so the read-out loop accounts for all collective talk variance.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH342 proposed a parameter-free check of the read-out loop gain. For a linear point process with branching ratio g , collective count variance at long windows exceeds the independent-agent value by $1/(1 - g)^2$. H111 compares this sum rule, with H67's g_{lag} , against the collective Fano ratio Φ of 10-min talk counts in 66 non-holdout units (35 periods). In regime III the rule closes: the pooled ratio $r_F = \Phi/\Phi_{\text{pred}}$ is 1.01 [0.94, 1.09] over 18 units, with no between-unit spread, and $\Phi - 1$ tracks g_{lag} (ρ 0.60). No unit reaches the kill ($r_F \geq 2$). In regime I, where $g_{\text{lag}} \approx 0$, talk co-varies $1.34 \times [1.18, 1.52]$ more than the loop allows. The holdout is not run.



Observables. Left: observed $\Phi(10 \text{ min})$ (per-call clock, all-present window, hour-block demeaned) against the parameter-free prediction from H67's g_{lag} , per unit with 95% CIs. Regime-III units (dark) lie on the line; regime-I units (light) sit above it at $\Phi_{\text{pred}} \approx 1$. Right: $\Phi(T)$ medians and IQRs. Regime III is flat from 5 to 30 min near the median prediction, so no slow field remains.

PHYSICAL MODEL

Linear multivariate Hawkes process on the call clock (model 09) with a fluctuation decomposition (model 14). Each agent has private talk noise of any form (clocked calls, own bursts). Reading couples agents through a cross-response matrix K with eigenvalue g on each room's collective mode. n_i is agent i 's talk count in a window, Φ the ratio of collective to summed private variance, N the present agents, n_b the room sizes. Φ divides out every private noise level.

$$C(T)/T \rightarrow (I - K)^{-1} \Sigma (I - K)^{-\top}, \quad \Phi(T) = \frac{\text{Var} \sum_i n_i}{\sum_i \text{Var} n_i}$$

$$\Phi_{\text{pred}}(g) = \frac{N(1 - g)^{-2}}{\sum_b [(1 - g)^{-2} + \frac{n_b - 1}{(1 + g/(n_b - 1))^2}]} \approx \frac{1}{(1 - g)^2}$$

$$r_F = \Phi(10 \text{ min}) / \Phi_{\text{pred}}(g_{\text{lag}})$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator, H67's read-out gain is the whole story of collective talk clustering in the current scaffold: a push is amplified by $1/(1 - g) \approx 1.15$ and the swarm's talk variance by $1/(1 - g)^2 \approx 1.32$, with nothing left over for a hidden field at 10–30 min. A monitor can use Φ on 10-min talk counts as a cheap, label-free check of the coupling gauge: Φ above $1/(1 - g_{\text{lag}})^2$ by more than ~ 0.3 flags a field or a coupling the read-out estimator misses. In regime-I-style scheduled chat, expect that excess ($\Phi \approx 1.3$ with $g_{\text{lag}} \approx 0$).

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
P1 regime III $r_F \in [0.8, 1.2]$ in $\geq 2/3$	$r_F \in 14/18$; pooled 1.01 [0.94, 1.09]	supported
K kill: $r_F \geq 2$ in $> 1/2$	0/18	not met
P2 untrimmed trimmed	$> 11/18$; gap +0.03 (raw +0.22)	failed
P3 excess rises with T	1/17	failed
P4 regime I $\Phi > 1.2$ in $\geq 2/3$	13/21; pooled r_F 1.34	failed narrowly
P5 $\rho(\Phi - 1, g_{\text{lag}}) \geq 0.3$	all 0.09; regime III 0.60	failed / III yes
P6 within $>$ cross room	10/12; cross > 0 in 0/12	mixed
P7 call clock lowers Φ	13/18	supported
NE43 bookends: trim flat, raw falls	-0.01, +0.08; +0.23 $\rightarrow$ -0.05	supported
NE42 Φ drops in #40	-0.04 vs predicted 0.41	failed

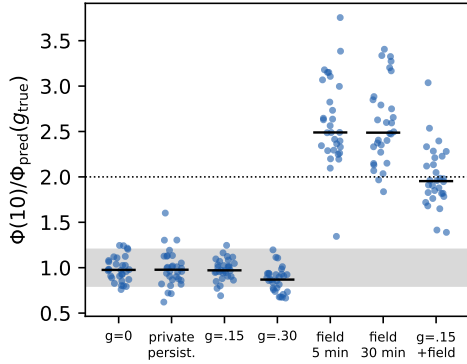
FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	2	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Per-unit CIs are ± 0.3 – 0.5 , so the HH's $\pm 20\%$ per unit is not resolvable; the pooled regime-III ratio carries the claim. A fast field ($\tau \leq 5$ min) and a missed coupling look alike in $\Phi(T)$, so the regime-I excess is unattributed. The per-call clock was adopted after the synthetic run showed a call-grid field (Amendment A1, before real data). The block-shift identity $F/F_{\text{shift}} = \Phi$ fails within 10% in half the units. Predictions were written knowing H67's g_{lag} .



Second figure. Synthetic validation on the real call grids of units 27, 41 and 51e (10 replicates per world). Coupling alone is recovered within $\pm 3\%$ at $g = 0.15$ and reads 5–20% low at $g = 0.30$. Private persistence leaves Φ at 1. Shared rate fields of 5 or 30 min give $\Phi/\Phi_{\text{pred}} = 2.3$ – 2.9 .

ROUND 2

Recompute Φ_{pred} in regime I with a next-chat-call gain; split the regime-I excess with a window-level in-flight placebo; run the frozen confirm after H67's.

- **Essence.** a closure test that says, per period, how much collective talk variance the measured coupling explains and how much is field.
- **H111-R1** Regime-I clock.
- **H111-R2** Fast-field separation.
- **H111-R3** NE42 #40.
- **H111-R4** Holdout.

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 d descriptive
G05 d descriptive
G06 ✓ supported
G07 d descriptive
G08 ✓ supported
G10 ~ mixed
G11 d descriptive
G12 d descriptive
+28 more periods: see the card.